

Tanmoy Sen Gupta

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SUMMARY

Software Engineer with expertise in backend development, distributed systems, and open-source tooling. Experienced in building high-performance data stores, developer libraries, and automation tools using Go, Python, and JavaScript. Passionate about crafting well-documented, maintainable software and contributing to the open-source community.

TECHNICAL SKILLS

Languages: Go (Golang), Python, JavaScript, TypeScript, PHP, Shell/Bash
Frameworks & Libraries: Flask, React, Node.js
Databases & Storage: MongoDB, JSON-based data stores, RESTful API design
DevOps & Cloud: Docker, Kubernetes (AKS, EKS, GKE), GitHub Actions, Kustomize
Tools & Platforms: Git, Homebrew, Linux, Makefile, Goreleaser
Concepts: Distributed Systems, Cryptography, Data Security, Open-Source Development, CLI Tools

OPEN-SOURCE PROJECTS

- wunderDB** | *Go, Docker, Kubernetes* | github.com/TanmoySG/wunderDB 2020 – Present
- Designed and built a JSON-based in-memory data store inspired by MongoDB with a RESTful API.
 - Developed `wdbctl`, a Homebrew-installable CLI tool for managing wunderDB instances.
 - Implemented Docker support and published versioned container images for seamless deployment.
 - Maintained Go client library (`wdb-go`) and interactive dashboard (`wunderDash`) for the platform.
- go-steps** | *Go* | github.com/TanmoySG/go-steps 2023 – Present
- Created a Go library enabling developers to define, execute, and manage functions as sequential step-chains.
 - Implemented conditional branching, configurable retry logic with error pattern matching, and sleep intervals.
 - Built built-in rollback functionality with `RollbackFn` for automatic cleanup on step failure.
 - Integrated zerolog-based structured logging with per-step progress tracking via `GoStepsCtx`.
- K-Deploy Action** | *GitHub Actions, JavaScript, Kubernetes, Kustomize* | github.com/TanmoySG/k-deploy-action 2022
- Built a GitHub Action to automate Kubernetes pod deployments across AKS, EKS, and GKE clusters.
 - Added support for Kustomize overlay management for environment-specific deployments.
 - Published to the GitHub Actions Marketplace enabling zero-config CI/CD Kubernetes deployments.
- logsmith & logsmith-monitor** | *Python, JavaScript* | github.com/TanmoySG/logsmith 2022
- Developed `logsmith`, a structured logging library for Python applications with configurable output formats.
 - Built `logsmith-monitor`, a centralized log aggregation and monitoring server for distributed services.
- hybrid-cryptography** | *Python, Jupyter Notebook* | github.com/TanmoySG/hybrid-cryptography 2021
- Researched and implemented advanced data security using AES, RSA, and Blowfish algorithms in a layered approach.
 - Designed a secure key-sharing protocol combining multiple cryptographic algorithms for enhanced data protection.
- Mycroft** | *Python, Flask, React* | github.com/TanmoySG/Mycroft 2020
- Built a full-stack web application for RSA-based encryption and decryption of sensitive data.
 - Integrated a Flask backend with a React frontend, exposing a RESTful API for cryptographic operations.

EDUCATION

[University Name] [City, Country]
Bachelor of Technology in Computer Science and Engineering 2018 – 2022

- Final Year Project: *hyENC3* – Advanced multi-layer hybrid cryptography system for securing sensitive data.

INTERESTS & COMMUNITY

Open-Source: Actively maintains multiple open-source Go and Python libraries used by the developer community.
Technical Writing: Publishes engineering articles and guides at blog.tanmoysg.com.
Apache Kafka: Completed 100 Days of Code challenge with Apache Kafka (Confluent).